NV-deCYPher submission report

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This document outlines our approach to the regression and classification tracks. We would like to thank the organisers for providing the data and putting this challenge together.

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1. Overall approach

We built our entries on pretrained molecular representation models, which consistently outperformed traditional descriptor + tree-based pipelines and non-pretrained graph neural networks across both tracks and the four scored CYP isoforms. We used two model families throughout: **Contrastive KERMT**, a pretrained molecular transformer, and **Uni-Mol2**, a 3D conformer-based model. For KERMT, we used both the original pretrained backbone and a variant with corrected padding behavior. Both model families were fine-tuned as multi-task regressors over all four isoforms simultaneously, together with parallel auxiliary endpoints, and then ensembled separately for each isoform.

While pretraining and multi-task supervision brought real and reproducible performance boosts, once both backbones were in place, further architectural work returned relatively little: the model size, head topology, hyperparameter sweeps had limited effect. And, surprisingly for us, so too was incorporating external datasets (more below). The two differentiators for us were the data treatment, and the calibration of our predictions.

We also used this opportunity to develop workflows that incorporate reasoning LLMs alongside expert molecular-based models, focusing our efforts on the classification track.

2. Labelled data sources

Datasets used

- **Challenge training set:** direct-inhibition pIC50 (the four scored endpoints), plus the single-concentration screen as log2 fold-change on four auxiliary heads.
- **Public CYP inhibition screens:** PubChem AID 1851, the Tox21 P450 qHTS screen, and the NCATS ADME CYP panel.

- **ToxCast:** NovaScreen (NVS) ADME CYP panel, invitroDB v3.3, EPA CompTox
- **Non-CYP auxiliaries:** the OpenADMET PXR challenge (pregnane X receptor induction) and the CarbonMangels substrate flags for CYP2C9, CYP2D6 and CYP3A4.

Datasets not used

- **ChEMBL.** We explored multiple ways of incorporating labelled CYP data, including treating censored values as one-sided bounds, non-binders pinned to a low value under a hinge loss, and only including similarity-restricted (Tanimoto) subsets close to the blind test set. All were neutral or harmful. Given the volume of ChEMBL data left out, careful data curation is a clear opportunity for improvement. Note that (unlabelled) ChEMBL molecules are nevertheless part of our pretraining corpus for KERMT.
- **Literature time-dependent-inhibition sets.** The public corpus spans multiple incompatible assay conventions across dozens of laboratories and would have needed to be harmonized to the challenge's shift definition. We also excluded datasets behind subscription paywalls.
- **Emax.** Present only where pIC50 is also present, so it adds no coverage.
- **Public data on the blind test molecules themselves.** We uncovered records for 43 of 750 blind test set molecules across all CYP and non-CYP endpoints, but did not use them.

3. Data preparation

3.1 Imputation of unmeasured direct inhibition pIC50 values

As described in the challenge description, the single-concentration screen was run as a full panel, while the 12-point DRC follow-up was carried out on a selected subset. The missing pIC50 values are therefore informatively missing, and the selection rule would appear to differ by isoform. For the CYP2D6 isoform in particular, the follow-up set was strongly truncated with the unmeasured population lying effectively entirely outside the support of the labelled values.

We exploited this by fitting a monotone relation from screen fold-change to pIC50 on the molecules that have both, extrapolating it to the unmeasured CYP2D6 molecules, and training on the result as a one-sided bound under a hinge loss rather than as a point label. The model is penalised for predicting more potent than the bound and is free below it.

This resulted in the single largest late-stage improvement of our predictions on CYP2D6 by a wide margin. We applied the identical construction to CYP1A2 and CYP2C9 but there was unfortunately no corresponding benefit.

3.2 A provenance artefact in the CYP3A4 classification labels

1249 CYP3A4 molecules carry a TDI pIC50 measurement with no corresponding direct-inhibition measurement; *all* 1249 read False under the *is_TDI* label, against a ~33% positive rate among the 2334 molecules carrying both arms. They also have no single-concentration data and occupy a disjoint registration-identifier block.

We excluded those molecules from classification training rather than relabelling them, on the grounds that we had no evidence the direct inhibition arm was ever run. Gating them out moved the CYP3A4 training prevalence from 21.7% back to 33.0%, against a rate of ~31.0% in the live portion of the blinded test set.

3.3 Parent-seed block structure

The blind set is organised as 75 chemical series of ten analogues around a parent compound. We recovered this structure by clustering on Morgan-fingerprint Tanimoto similarities.

For the classification track it supplied a per-series prior. Reading off the parent labels for the analogues as coarse grained predictions, we used our models as fine-grained discriminators within the series. This gave the majority of the positive predictions across the TDI-positive parent series. We also applied the same approach to one series whose parent did not clear the screen and was therefore not a TDI-positive hit, leading to five further TDI positives. Together, this approach produced our largest single gain on CYP2D6.

For the regression track, the inferred block structure proved less useful – the prior signal is mostly recovered by our models and the main benefit is a mutual validation of the clustering approach and our modelling efforts.

4. Model architectures

For KERMT (GitHub, paper), we pretrained our own backbones from scratch for 100 epochs, following the same hybrid architecture and self-supervised training recipe as the publicly released model. The pretraining corpus consisted of the original KERMT corpus (ZINC15/ChEMBL, Biogen, OpenADMET-ExpansionRX, and ChEMBL multitask training

splits), supplemented with public CYP inhibition datasets (PubChem AID 1851 and CarbonMangels) and the training molecules from the CYP challenge, totaling approximately 10.06 million molecules. Pretraining was performed on SMILES alone, without using any experimental labels, and only the training splits of the additional datasets were included. We also pretrained a variant that fixes a padding issue in the encoder, which previously caused a molecule's embedding to depend on other molecules in the same batch. Both the original and padding-corrected backbones were used in our final ensemble. We also excluded the blind test molecules from pretraining to keep the approach purely inductive. It is likely that a transductive approach would have helped, but retraining a model for every separate investigation is highly impractical and so we excluded the approach to reflect that.

For Uni-Mol2 we fixed 25 precomputed conformers per molecule and averaged predictions across them at inference. Post-hoc analysis of this test-time ensembling found very small impact on predictions (typically Spearman $\rho > 0.99$). We experimented with all published model sizes but in limited testing found no obvious benefit in scaling up, hence we used the smallest 84M parameter variant throughout, together with a single 1.1B seeded variant trained as part of the initial testing.

5. Training

- **Objective.** A mix of MAE and MSE on observed cells. Entries carrying an imputed bound instead contribute a one-sided hinge.
- **Order supervision.** An auxiliary pairwise ranking term over molecule pairs in which only one member has a measured pIC50 but both have a screen or external source reading, so the screen contributes ordering information where it cannot contribute a value. Weighted well below the primary loss at $\lambda = 0.1$.
- **Cross-validation.** Five folds with up to five random seeds, with the 25 resulting models averaged.

6. Ensembling

Per isoform, the two backbone families were combined as a weighted blend, with weights set from each member's validation loss (random split was used to create train/validation splits) and its correlation with the incumbent column, rather than by grid search.

Models trained with order supervision produce a different value distribution from those trained with value losses alone, so blending them directly distorts the marginal. Instead, we blended those in rank space and then restored the value distribution by quantile matching against the MAE-trained reference, which preserves the ordering gain without the distributional side-effect.

For the classification track, CYP2D6 predictions were combined with the parent-series prior described above. For CYP3A4, the ensemble combined the KERMT and Uni-Mol2 predictors with a post-trained SFT Nemotron 3.5 Lightning checkpoint and a feature applying the label definition to our predicted pIC50s. Each component's weight was allocated by its marginal contribution to the blend rather than its standalone performance.

7. Calibration

For the regression track, we perform calibration in two final steps before submission.

First, each isoform's predictions are mapped to a fixed mean and standard deviation, which we determine empirically from live leaderboard submissions. This is particularly crucial for 2D6, as most participants would have realized very early on from negative R^2 metric scores. The open risk here is overfitting to the live leaderboard.

Second, the lowest pIC50 values in the training set molecules are ~2.0, and the wide confidence intervals of ~1.0 in that low-potency pIC50 range suggests that applying a prediction floor of ~3.0 pIC50 is nearly risk-free, given the ST-RAE metric. Applying this floor improved all four isoforms.

For the classification task, we use the leaderboard's reported precision and recall to estimate the number of positives in the blind set: ~216 for CYP3A4, ~46 for CYP2D6. For CYP3A4 we called 226, the count that maximised expected MCC given how well our scores separate positives from negatives, measured as AUC. For CYP2D6 we called 34: 29 coming from seven TDI-positive parents as described under the parent-seed block structure, plus five from one unassayed block. We stopped there because molecules outside those blocks are so rarely positive that calling them adds more false positives than true ones.

8. Agent-assisted analysis

A substantial fraction of the analysis in this work from experiment design, diagnostics, leaderboard interpretation, and record-keeping, was carried out with LLM coding agents.

One factor limiting its overall usefulness, though, was the large distribution shift between training and the blind test sets. This blocked us from deploying fully autonomous exploration and iterative refinement agents.

Another use of our agent was in sourcing ideas from prior competition solutions (e.g. from publicly declared Kaggle winning solutions). Honestly assessed, with the caveat that the token budget for this task was not infinite, the yield here was low. Only one idea was adopted, namely the pairwise order supervision described above; most of the remainder was either specific to the original task or already standard practice.

9. Reasoning LLMs as TDI predictors

We investigated how best to incorporate reasoning LLMs alongside predictive models built directly on molecular structure. We focused on time-dependent inhibition, which poses a binary classification question rather than requiring a continuous regression output: is a given molecule a TDI inhibitor? We tested more than ten frontier models with varying amounts of task context and molecular information. Zero-shot performance was generally close to zero. Adding predictive models (both KERMT and other community-developed models) as tools in a harness substantially improved performance, but the resulting predictions suggested that the LLMs did little beyond averaging the tool outputs. Our classification entry using this approach was outperformed by the straightforward predictor ensembles that followed.

Next, we tested whether post-training could develop task-specific reasoning and improve how the models used the predictors. We trained models in the Nemotron family with the predictors available as callable tools. The task was framed as a multi-turn interaction in which the model chose which tool to call next, within a fixed step budget. Synthetic traces were filtered for correct answers without label leakage and used for supervised fine-tuning (SFT), followed by GRPO. Starting GRPO directly from the zero-shot models proved hard to train and collapsed early, making the SFT warm start crucial in our experiments.

We studied Nemotron 3.5 Lightning (HuggingFace) in detail, using synthetic traces generated by a larger OSS teacher model as a warm start for SFT. SFT substantially improved CYP3A4 discrimination over zero-shot performance, with GRPO providing a further gain. CYP2D6 performance, however, deteriorated at each stage. This was consistent with our broader finding that series structure was particularly informative for CYP2D6 and may be difficult to exploit when reasoning about molecules individually.

Our final CYP3A4 predictions are a rank-space blend of the KERMT and Uni-Mol2 models, the SFT checkpoint of Nemotron 3.5 Lightning, and a feature that applies the label definition directly to our predicted pIC50s rather than learning it. Ensemble weights were set by how decorrelated each component was from the others, so components earned their weight by ranking molecules differently rather than by being strong alone. GRPO improved standalone MCC and F1 but correlated strongly with the SFT checkpoints and added no ensemble benefit. Structure-based prediction and reasoning over tool outputs have different strengths, and the strongest entry combined them rather than relying on either as a standalone predictor.

Example: how post-training can improve domain-specific reasoning

To illustrate how post-training could improve domain-specific reasoning, we offer an example of a query answered by the same model before and after SFT, in the same harness with the same ten tools. The molecule is CCN(Cc1ccc2c(c1)OCO2)C(=O)NCc1nnc2n1CCCCC2, a CYP3A4 TDI-positive carrying a benzodioxole. Both models answer Yes; only one does so for a chemical reason.

The model is given no molecule-specific numbers in the prompt: every quantity it reasons over must be fetched by calling a tool. The three structure-based predictors — **KERMT, Chemprop and Uni-Mol2** — are reached through the model argument of the first three tools, so the model chooses which predictors to consult and whether to consult all of them. Ten tools are available:

- predict_tdi(smiles, isoform, model) — out-of-fold TDI probability, and its percentile among training molecules
- predict_potency(smiles, isoform, arm) — predicted direct or pre-incubated pIC50, and the implied shift
- tool_reliability(isoform, model) — that predictor's measured out-of-fold ROC-AUC, RMSE and r on that isoform
- nearest_neighbours — up to 25 labelled training molecules by Tanimoto similarity, with their measured results
- structural_alerts — bioactivation SMARTS matched by the molecule
- alert_enrichment — measured TDI rate with versus without each alert
- base_rate — TDI rate overall and by direct-pIC50 bin
- descriptors — RDKit physicochemical properties
- screen_hit_likelihood — likelihood the molecule was a single-concentration screen hit
- isoform_literature — known inactivation mechanisms for the isoform

Answer A corresponds to Yes, answer B to No. The prompt below is the CYP3A4 one; the CYP2D6 prompt is the same except for its isoform-specific label rule, mechanism paragraph and prevalence figure.

You are a drug-metabolism expert answering a CYP time-dependent-inhibition question. You have tools that query our in-house training data: structure-based model predictions and their MEASURED out-of-fold reliability, nearest labelled training neighbours (you choose how many, up to 25), structural alerts, RDKit descriptors and base rates. No numbers are given to you in advance - call the tools you actually need, and call them again with different arguments if the first answers are uninformative. Do not call tools you do not need. When you have enough evidence, stop calling tools and give a concise expert analysis (100-200 words, plain prose, no headers) weighing what you found. Never name a drug or claim to recognise the compound. Be honest about uncertainty. End with a final line exactly "Answer: X" where X is A or B.

Is this molecule a time-dependent inhibitor (TDI) of the cytochrome P450 enzyme CYP3A4?

Official label rule. Let direct = pIC50 without pre-incubation and preinc = pIC50 after pre-incubation, shift = preinc - direct.
- positive (Yes): direct > 4 AND shift > 0.301 (more than a 2-fold drop in IC50);
- inferred positive (Yes): direct < 4 but preinc > 4.301;
- negative (No): direct > 4 and shift <= 0.301;
- assigned negative (No): both arms below 4.
The scored positive class is positive + inferred positive.

Mechanism, CYP3A4: a large, flexible active site (about 1,400 A^3) that accepts bulky and lipophilic substrates and can hold more than one ligand. Time-dependent inhibition of CYP3A4 is usually mechanism-based: the enzyme oxidises the compound to a reactive species that alkylates the heme or the apoprotein, or forms a quasi-irreversible metabolic-intermediate (MI) complex. Classic motifs are methylenedioxyphenyl (carbene MI complex), furans and thiophenes (epoxide / S-oxide), terminal alkynes (heme alkylation), and secondary or tertiary alkylamines (nitroso MI complex). Direct inhibition and time-dependent inhibition are different properties: in this data set molecules with essentially no direct inhibition (pIC50 below 4) are rarely TDI-positive, but above that threshold the TDI rate is roughly flat with direct potency. You can query the base rates with the tools.

SMILES: [molecule]

Options:
A) Yes
B) No

Two further requirements. (1) Across the whole screened population about 29 in 100 molecules are TDI-positive for CYP3A4, so answer Yes only for a molecule you would rank in roughly the top 29% of screened compounds on this evidence - judge it against that population, not in isolation. (2) Before the final line, give your calibrated probability on its own line, exactly "P(Yes) = 0.xxx" to three decimals.

Zero-shot model:

Overall TDI positive fraction 0.327, which is about 33% of screened molecules. Our molecule's percentile among training is 64, which is above 29% threshold. So likely positive.

Now we need to produce final answer: probability 0.xxx. We have tdi_probability 0.392 from predict_tdi, percentile 64. Also base_rate 0.327. So probability maybe around 0.4? But need calibrated probability on its own line "P(Yes) = 0.xxx". Use maybe 0.40? But need three decimals. Let's output 0.40? Actually 0.400. But maybe 0.40 is 0.400. Use 0.400.

But also need answer A or B. Since probability >0.29, answer Yes => A.

Thus final: P(Yes) = 0.400, Answer: A.

Let's produce.

P(Yes) = 0.400
Answer: A

The benzodioxole is never mentioned. No mechanism, no alert query, no reliability check: one probability is retrieved, a percentile is compared against the stated prevalence, and the rest of the output goes on decimal formatting. The answer is correct, but it is the output of a thresholding procedure over a tool's rank.

Post-trained model:

Let me weigh the evidence:

1. Structural alert: methylenedioxyphenyl (benzodioxole) — classic CYP3A4 mechanism-based inactivation motif (carbene MI complex). Measured TDI rate with this alert: 68.2% vs 32.4% without (lift 2.08). Strong evidence for Yes.

2. Model TDI probabilities: kermt 0.392, chemprop 0.463, unimol2 0.783. Mixed. Ensemble AUC 0.789. The unimol2 high probability (90th percentile) is notable.

3. Potency predictions: direct pIC50 ~4.1-4.7 (above 4), shift ~0.35-0.49 (above 0.301 threshold). All three models predict direct > 4 and shift > 0.301. But shift RMSE is 0.35-0.36, which is at/above the 0.301 threshold — so shift predictions are noisy. Still, all three models agree on a positive shift.

4. Nearest neighbours: low similarity (max 0.372), 30% positive. Not very informative.

5. Base rate: overall TDI rate ~32-33%. With methylenedioxyphenyl alert, 68%.

The methylenedioxyphenyl group is the key here. Benzodioxoles are classic CYP3A4 inactivators via carbene MI complex formation. The alert enrichment is strong (68% TDI rate). The models' shift predictions all exceed the threshold, and direct pIC50 is predicted above 4. The TDI probability ensemble: average ~0.55, but unimol2 is high.

Weighing: the structural alert with measured 68% TDI rate is the strongest evidence. The models' shift predictions all agree on >0.301, and direct > 4. The nearest neighbours are not very similar and mostly negative, but they're at low similarity so they carry little weight.

I lean toward Yes (A). The methylenedioxyphenyl alert is the decisive factor here. Even though the TDI probability models are mixed, the measured enrichment of the alert is compelling. The shift predictions from all three models exceed the threshold, and direct potency is predicted above 4.

Answer: A.

The decisive evidence here is the methylenedioxyphenyl (benzodioxole) group, a classic CYP3A4 mechanism-based-inactivation motif that forms a carbene metabolic-intermediate complex with the heme iron. This alert is strongly enriched in the training data: molecules bearing it are TDI-positive 68% of the time versus 32% without (lift 2.08, n=22). All three models predict direct pIC50 above 4 (4.1–4.7) and a pre-incubation shift of 0.35–0.49, i.e. above the 0.301 threshold, so the label rule would be met. The TDI-probability models are mixed (0.39–0.78), but the ensemble AUC is 0.79, and the unimol2 model (90th percentile) is the most bullish. Nearest neighbours are too dissimilar (max 0.37) to override this. The main caveat is that shift RMSE (~0.35) is comparable to the threshold, so the shift prediction is not certain; however, the structural alert enrichment plus unanimous shift predictions on all three models tip the balance toward a positive call.

Answer: A

The mechanism and the tools are used together rather than in sequence: the benzodioxole supplies the hypothesis, the alert-enrichment tool converts it into measured evidence, the reliability tool decides how much the three disagreeing predictors are allowed to count, and the nearest neighbours are consulted and then explicitly discounted at a maximum similarity of 0.37. Post-training aims to teach the model how to use the available tools together with its prior knowledge.

Note that both traces above are from a smaller Nemotron 3 Nano 4B model (HuggingFace), trained on the same teacher traces, tool set and harness as the Nemotron 3.5 Lightning model — the zero-shot run is the base model and the post-trained run is the same base model after SFT. This model was not part of the final ensemble and this is a single illustrative case rather than a quantitative result.